

Sessions 4 & 5: Macro at risk Heteroskedasticity, outliers and quantile regressions

Modern Macroeconometrics

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OVERVIEW OF SESSIONS 4 & 5

› Session 4 – Heteroskedasticity and outliers

- › t -distributed errors
- › Outlier-augmented stochastic volatility (SV)

› Session 5 – The macroeconomy at risk

- › Bayesian quantile regression
 - › From quantiles to predictive densities
- ## › Tail risk forecast evaluation

Session 4

Heteroskedasticity and outliers: Why stochastic volatility?

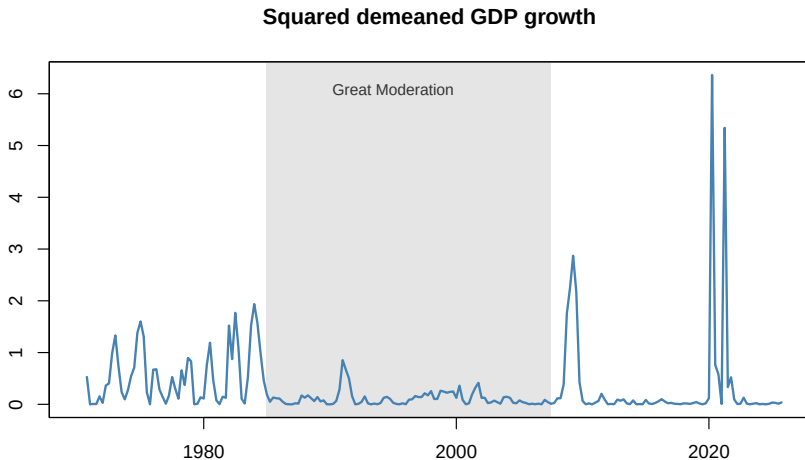
RECAP FROM SESSION 3 AND WHAT IS NEXT

- In Session 3 we introduced the mechanics of **stochastic volatility (SV)**, with the log variance h_t following an AR(1) process
- In this session we take a look at a few practical considerations of SV
 - Why SV is crucial in any applied macroeconometric toolkit nowadays?
 - How do we cope with extreme **outliers**, such as those observed during COVID?

WHY MODEL TIME-VARYING VOLATILITY?

- Similar to financial time series, we also tend to observe volatility clustering in macro series
- Homoskedastic errors may result in **misspecification**
 - **Great Moderation**: GDP and inflation volatility fell sharply from the mid-1980s and stabilised
 - **GFC** and **COVID-19**: Extreme, abrupt swings inflate volatility; is this surge in volatility persistent or transitory?
- Ignoring heteroskedasticity may result in **misstated forecast uncertainty** with intervals too narrow in turbulent times and too wide in tranquil times

WHY MODEL TIME-VARYING VOLATILITY?



DRIFTING VARIANCES AND OUTLIERS

- **Gradually moving heteroskedasticity:** The variance drifts slowly over time, leading to volatility clustering
- **Occasional outliers:** Isolated large deviations (COVID or GFC), captured by **heavy-tailed errors** or **explicit outlier components**
- The two differ in nature and require different modelling assumptions
- SV (h_t as an AR(1) process) assumes **persistence**, capturing volatility clustering
- But an outlier is typically just an on-off spike
- In practice (especially since COVID), an SV model that can accommodate outliers is needed

A BRIEF DIGRESSION: GARCH

- Alternative to SV, the workhorse for time-varying volatility are **ARCH** and its generalisation **GARCH** (see Engle 1982, *ECTA* and Bollerslev 1986, *JoE*)
- A GARCH(1,1) models the conditional variance as a deterministic function:

$$\sigma_t^2 = \alpha + \beta \varepsilon_{t-1}^2 + \gamma \sigma_{t-1}^2$$

- **Observation-driven**: Today's variance is pinned down by yesterday's squared shock and variance; there is **no separate volatility innovation** (i.e., state innovations)
- Captures **volatility clustering** and fat tails and is cheap to estimate by (quasi) maximum likelihood

SV vs GARCH

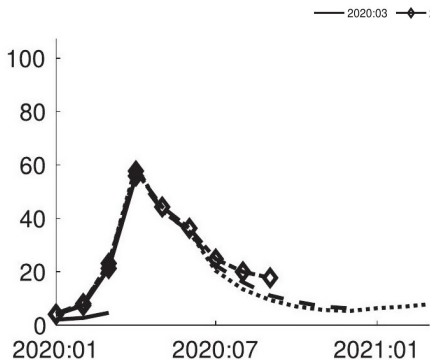
- › GARCH reacts “**mechanically**” to the last shock
- › SV is **stochastic**
- › **SV** (Session 3): The variance has its **own latent shock**, $y_t = \exp(h_t/2) \varepsilon_t$ with h_t a stochastic AR(1) process: $h_t = \mu_h + \phi_h (h_{t-1} - \mu_h) + \eta_{h,t}$, $\eta_{h,t} \sim \mathcal{N}(0, \omega_h)$
- › **Why SV for macro**: It captures slow, smooth changes in volatility, delivers realistic density forecasts, and can be implemented easily in our **modular Bayesian toolkit**

Extensions to SV and alternatives

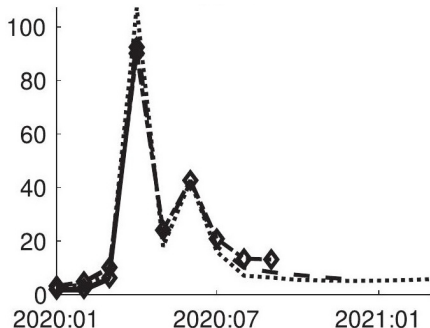
WHEN SV ALONE IS NOT ENOUGH

- › A single extreme observation (COVID 2020Q2) may force SV to explain it by inflating the **whole volatility path**, keeping it elevated long after the shock
- › In addition, a TVP model may instead absorb it as **spurious parameter shifts**
- › We want to treat such episodes as **transitory outliers**, not permanent changes in volatility paths
- › Two alternatives: (1) heavy-tailed errors or (2) explicit outlier components

COVID AS A VOLATILITY EPISODE: SV vs OUTLIER-AUGMENTED SV



SV: Volatility spikes in April 2020 and remains elevated; treats Covid as a **more persistent** shift



SVO-t: Volatility spikes sharply but quickly reverts; identifies Covid as a **transitory** outlier

(0) AD HOC DOWN-WEIGHTING

- Simplest practical fix: **Down-weight or “dummy out”** the pandemic observations
- Transparent, cheap and typically robust
- But the researcher chooses what counts as an outlier
- Lenza & Primiceri (2022, *JAЕ*) state that this convenience comes at a cost:

“The ad hoc strategy of dropping these observations may be acceptable for the purpose of parameter estimation. However, disregarding these recent data is inappropriate for forecasting the future evolution of the economy, because it may underestimate uncertainty.”

(1) HEAVY-TAILED ERRORS: STUDENT- t

- › Keep the SV variance but give the errors **heavier tails**
- › The Student- t is a **scale mixture of Normals**:

$$y_t = \mathbf{x}_t' \boldsymbol{\beta}_t + \exp(h_t/2)/\tau_t \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1), \quad \tau_t^2 \sim \mathcal{G}(\nu/2, \nu/2)$$

- › An outlier receives a small τ_t (i.e., large variance) and is **down-weighted**
- › Conditional on τ_t the model is Gaussian
- › A **trivial extra Gibbs block** to learn ν ($\nu < 5$: heavy tails; $\nu > 30$: approximately Gaussian)
- › Combined with SV this is the **SV- t** model; with constant variance this reduces to Student- t errors
- › Chan (2020, *JBES*) uses t -errors to absorb outliers in large (conjugate) BVARs

(2) EXPLICIT OUTLIER COMPONENTS

- Add an **outlier scale** o_t that multiplies the shock standard deviation:

$$y_t = x_t' \beta_t + \exp(h_t/2) o_t \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1),$$

$$o_t = \begin{cases} 1 & \text{with probability } 1 - \pi \quad (\text{no outlier}) \\ \sim \mathcal{U}(2, 20) & \text{with probability } \pi \quad (\text{outlier}) \end{cases}$$

- The posterior $p(o_t \mid \bullet)$ **identifies** which periods are outliers
- Carriero, Clark, Marcellino & Mertens (2024, *ReStat*): SV with outliers treats COVID as an **outlier-contaminated volatility** episode
- A Student- t extension (SVO- t) also possible

Stylistic application: GDP growth on NFCI

Single-covariate GaR regression under different error models

ONE-YEAR-AHEAD GROWTH TARGET

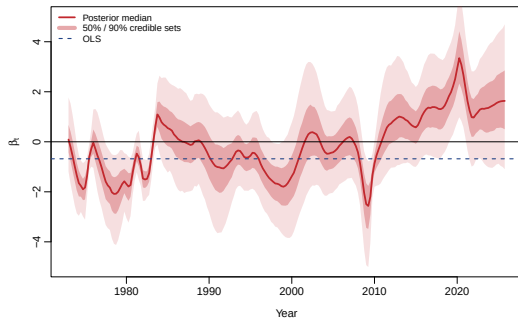
- Following Adrian, Boyarchenko & Giannone (2019, *AER*), the target is the **annualised average growth over the next four quarters**:

$$y_{t+h} = \frac{4}{h} \sum_{j=1}^h \Delta y_{t+j}, \quad h = 4$$

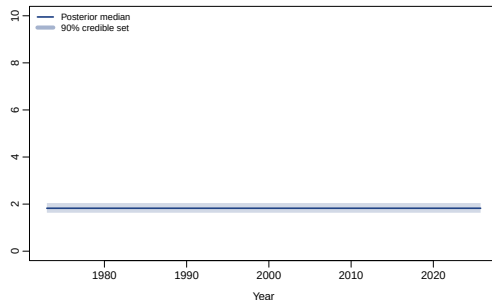
- Rolling / overlapping windows: consecutive targets share three quarters $\Rightarrow$ a smoother series and serially correlated observations
- We then regress it on **current** financial conditions: $y_{t+h} = \alpha + \beta \text{NFCI}_t + \varepsilon_t$

HOMOSKEDASTIC ERRORS: $\sigma_t = \sigma$

TVP coefficient on the NFCI

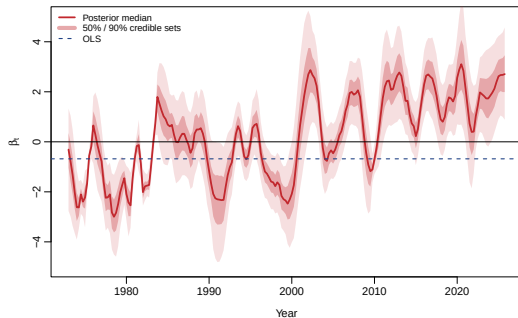


Estimated volatility σ_t

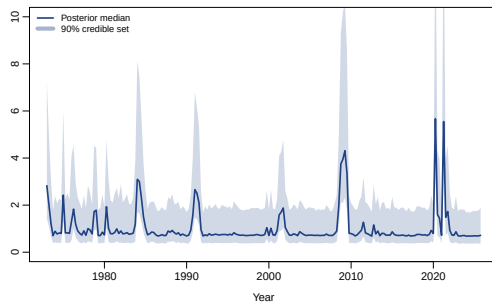


t -DISTRIBUTED ERRORS: $\sigma_t = \sigma / \tau_t$

TVP coefficient on the NFCI

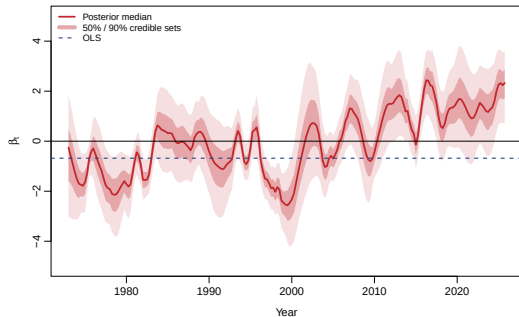


Estimated volatility σ_t

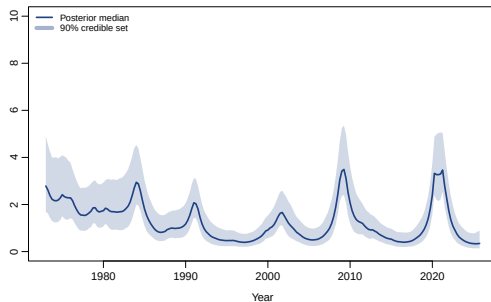


$$\text{SV: } \sigma_t = \exp(h_t/2)$$

TVP coefficient on the NFCI

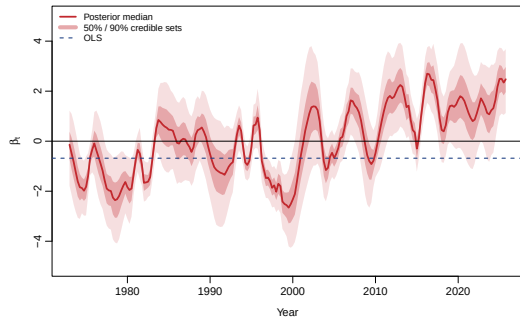


Estimated volatility σ_t

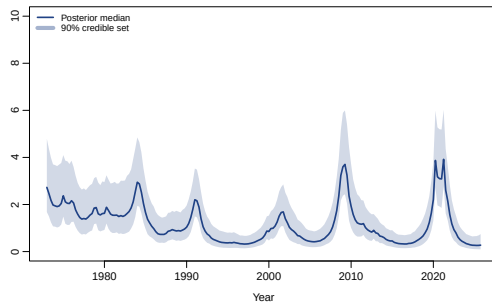


t -DISTRIBUTED SV: $\sigma_t = \exp(h_t/2)/\tau_t$

TVP coefficient on the NFCI

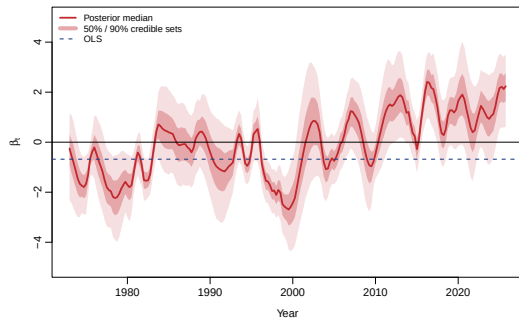


Estimated volatility σ_t

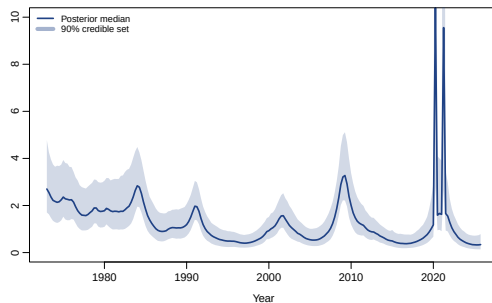


OUTLIER SV: $\sigma_t = \exp(h_t/2)o_t$

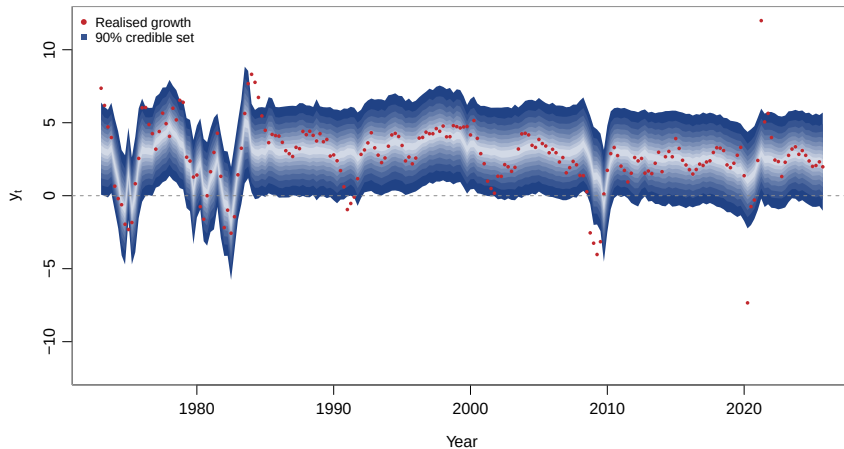
TVP coefficient on the NFCI



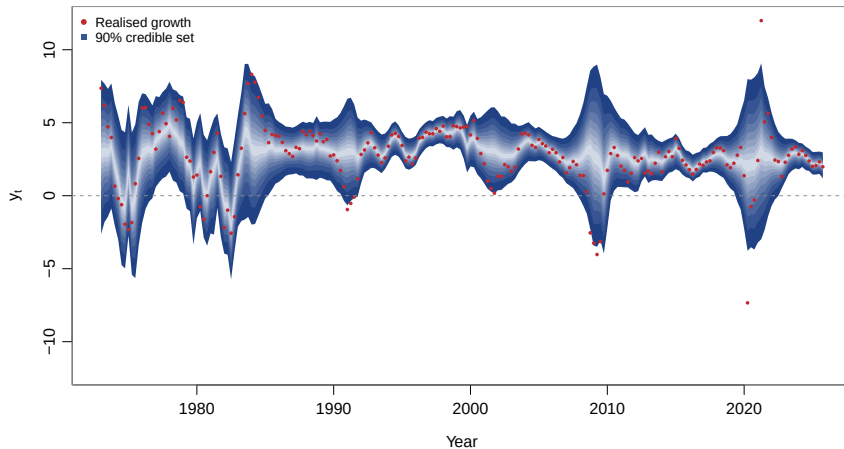
Estimated volatility σ_t



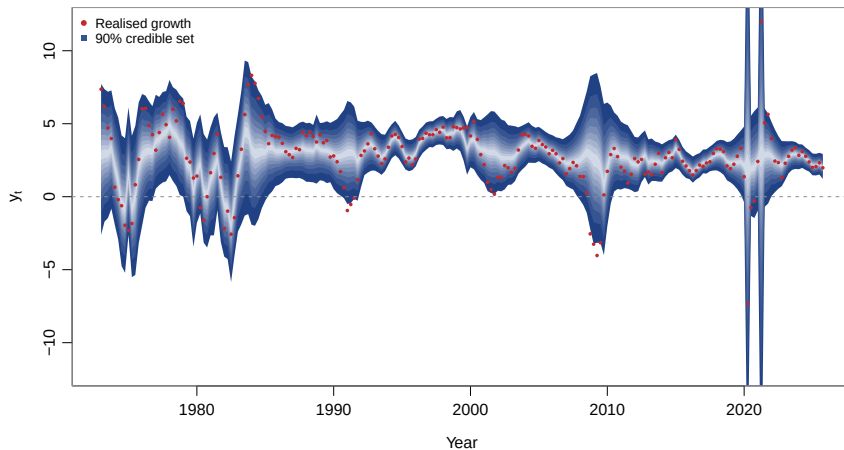
PREDICTIVE FIT: HOMOSKEDASTIC ERRORS



PREDICTIVE FIT: SV



PREDICTIVE FIT: OUTLIER SV



KEY TAKEAWAYS FROM SESSION 4

- Constant-variance models are misspecified for macro data
- **SV vs GARCH:** GARCH assumes that the variance is a deterministic function of past shocks; SV assumes it has its own latent process
- SV is a state space model; straightforward to incorporate in our Bayesian toolkit
- **Occasional outliers need their own treatment:** Heavy-tailed (t) errors, explicit outlier components, or down-weighting keep extreme episodes from distorting volatility and coefficients
- **Why it matters:** Credible density forecasts (intervals widen in turbulent times, narrow in tranquil ones) and unbiased structural inference
- The current/practical workhorse is **SV augmented with outlier robustness;** essential after COVID

Session 5

Quantile regressions, predictive densities, and tail-risk evaluation

From mean regression to quantile regression

LIMITATIONS OF MEAN REGRESSION

- › Standard regression models the **conditional mean** $\mathbb{E}[y_t|x_t]$
- › For symmetric, well-behaved distributions, the mean is a sufficient summary
- › In macroeconomics and finance, distributions are often **skewed and fat-tailed**:
 - › GDP growth: Skewed to the left (recessions are sharper than expansions)
 - › Asymmetric consumer price inflation and deflation risk
 - › Asset returns: excess kurtosis, negative skewness (leverage effect)
- › Policymakers demand more than the mean:
 - › “What is the probability of negative GDP growth?”
 - › “What is the probability of inflation being (substantially) above/below target?”
- › **Quantile regressions** directly models the τ th conditional quantile $Q_\tau(y_t|x_t)$

A QUANTILE REGRESSION (QR)

- › Koenker & Bassett (1978, *ECTA*) estimate the τ th conditional quantile $Q_\tau(y_t|x_t) = x_t'\beta_\tau$ by minimising:

$$\min_{\beta_\tau} \sum_{t=1}^T \rho_\tau(y_t - x_t'\beta_\tau)$$

with loss $\rho_\tau(u) = u [\tau - \mathbf{1}(u < 0)]$

- › **Each quantile** $\tau \in (0, 1)$ has its own coefficient vector β_τ
- › The loss function asymmetrically penalises positive and negative residuals
- › Median regression ($\tau = 0.5$) minimises L_1 loss $\sum_t |y_t - x_t'\beta_{0.5}|$
- › Standard QR gives estimates/predictions for each τ , not a full predictive distribution y_{t+h}

A BAYESIAN QUANTILE REGRESSION

- ▶ Yu & Moyeed (2001, *SPL*) write the QR loss as an **asymmetric Laplace (AL) likelihood**
- ▶ “A natural and effective way” to allow for Bayesian inference in a QR:

$$p(y_t | x_t, \beta_\tau, \sigma_\tau^2) = \frac{\tau(1 - \tau)}{\sigma_\tau^2} \exp \left\{ -\frac{\rho_\tau(y_t - x_t' \beta_\tau)}{\sigma_\tau^2} \right\}$$

- ▶ The $AL(\sigma_\tau^2)$ is asymmetric, with τ defining the degree of skewness
 - ▶ $\tau > 0.5$: Positive residuals penalised heavier; **left**-skewed (heavier left tail)
 - ▶ $\tau < 0.5$: Negative residuals penalised more; **right**-skewed (heavier right tail)
 - ▶ $\tau = 0.5$: Symmetric distribution

POSTERIOR AND DATA AUGMENTATION FOR QR

- Combine the AL likelihood with a prior $p(\beta_\tau)$ to obtain the posterior:

$$p(\beta_\tau | \mathbf{y}, \{\mathbf{x}_t\}_{t=1}^T) \propto \underbrace{\prod_{t=1}^T p(y_t | \mathbf{x}_t, \beta_\tau, \sigma_\tau^2)}_{\text{Likelihood}} \times \underbrace{p(\beta_\tau)}_{\text{Prior}}$$

- Kozumi & Kobayashi (2011, *JSCS*) show the AL likelihood has a **normal-exponential mixture** representation, allowing data augmentation:

$$y_t = \mathbf{x}_t' \beta_\tau + \theta_\tau v_t + \psi_\tau \sigma_\tau \sqrt{v_t} z_t, \quad v_t \sim \mathcal{E}(\sigma_\tau^2), \quad z_t \sim \mathcal{N}(0, 1)$$

where $\theta_\tau = \frac{1-2\tau}{\tau(1-\tau)}$ and $\psi_\tau^2 = \frac{2}{\tau(1-\tau)}$; all parameters $(\beta_\tau, \sigma_\tau^2, \theta_\tau, \psi_\tau)$ are **quantile-specific**, estimated/specified separately for each τ

POSTERIOR AND DATA AUGMENTATION FOR QR (CONT.)

- › Conditional on the auxiliary variable v_t this results in a linear regression for each τ

$$\tilde{y}_t | v_t \sim \mathcal{N}\left(\tilde{\mathbf{x}}_t' \boldsymbol{\beta}_\tau, \sigma_\tau^2\right), \quad \tilde{y}_t = \frac{y_t - \theta_\tau v_t}{\psi_\tau \sqrt{v_t}}, \quad \tilde{\mathbf{x}}_t = \frac{\mathbf{x}_t}{\psi_\tau \sqrt{v_t}}$$

- › Gibbs sampler iterates over
 1. $\boldsymbol{\beta}_\tau | \bullet$: Gaussian posterior (standard form)
 2. $\sigma_\tau^2 | \bullet$: Inverse Gamma (standard form)
 3. $v_t | \bullet$: Generalised inverse Gaussian (GIG), for each t
- › **Shrinkage and regularisation**: In high-dimensional settings (many predictors) the usual shrinkage priors (e.g., horseshoe, SSVS) can be used (see Session 2)

A FEW QR-SPECIFIC PRACTICAL CONSIDERATIONS

- › Estimate a **separate** regression for each quantile level $\tau \in (0, 1)$
- › **Quantile crossing**: Separately estimated quantiles need not be monotone in τ , so a crossed quantile function is not a valid CDF; post-processing fixes exist but are ad hoc
- › **Extreme quantiles**: At $\tau = 0.05$ or 0.95 the check loss is driven by relatively few observations, so estimates are noisy and credible intervals wide
- › **Choice of the τ -grid**: Too few levels ($L < 5$) give a coarse density fit; too many add noise in the sparse tails and make crossing more likely

PRODUCING PREDICTIVE DENSITIES WITH QRs

A TWO-STEP APPROACH

- ▶ **Step 1:** Estimate QR separately at each $\tau \in \{0.05, 0.10, \dots, 0.95\}$

$$\widehat{Q}_\tau(y_t|x_t) = x_t' \widehat{\beta}_\tau$$

- ▶ **Step 2:** Link the estimates together by fitting a parametric distribution to the L estimated quantiles
 - ▶ Find parameters $(\mu, \sigma, \alpha, \nu)$ of a skewed- t whose theoretical quantile function matches $\{\widehat{Q}_{\tau_k}\}_{k=1}^L$
 - ▶ The fitted skewed- t then serves as the predictive density $\widehat{p}(y_t|x_t)$
- ▶ Used in Adrian, Boyarchenko & Giannone (2019, *AER*)

RATIONALE FOR THE SKEWED- t

- › The skewed- t is a simple parametric family that keeps the features that matter for macro risk
- › **Asymmetry:** Skewness parameter α lets the left tail be fatter than the right
- › **Heavy tails:** Degrees of freedom parameter ν controls tail weight (independently of skewness)
- › **Parsimonious and tractable:** Four parameters only to match L quantiles, results in a closed-form CDF and quantile function

LIMITATIONS OF THE TWO-STEP APPROACH

- › **No coherence / quantile crossing:** The L QRs are estimated independently, so nothing prevents $\widehat{Q}_{\tau_1} > \widehat{Q}_{\tau_2}$ for $\tau_1 < \tau_2$
- › **Unimodal by construction:** The skewed- t can only shift mass left or right, it cannot represent a **bimodal** distribution
- › Mitchell, Poon & Zhu (2024, *IAE*) instead recover the density by **nonparametric interpolation** of the quantile function
- › They find that, under tight financial conditions, the GDP-growth distribution is **bimodal** (a recession cluster and an expansion cluster)

Stylistic application: GDP growth on NFCI

Single-covariate GaR-QR regression

GROWTH AT RISK: RECALL THE MAIN STORY

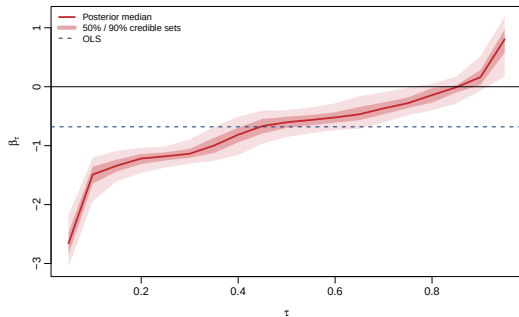
- **Growth-at-Risk (GaR):** Left tail (i.e., for small τ) of the predictive distribution of future GDP growth

$$\text{GaR}_\tau = Q_\tau(y_{T+h}|\mathbf{x}_T)$$

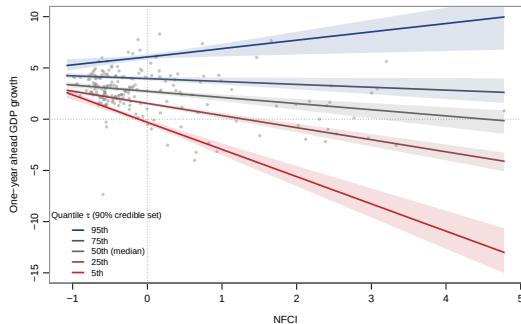
- Recall: Key empirical result from ABG (2019, *AER*):
 - **Financial conditions are asymmetrically informative:** they strongly predict the lower quantile of GDP growth, but have little effect on the median
 - This asymmetry is **invisible** to standard mean-based models
 - Tight financial conditions $\Rightarrow$ sharp deterioration in GaR; loose conditions $\Rightarrow$ moderate upside
- Same **one-year ahead average GDP-growth** target as the SV application, now via **quantile regression** on the **NFCI**

QRs ACROSS DIFFERENT τ

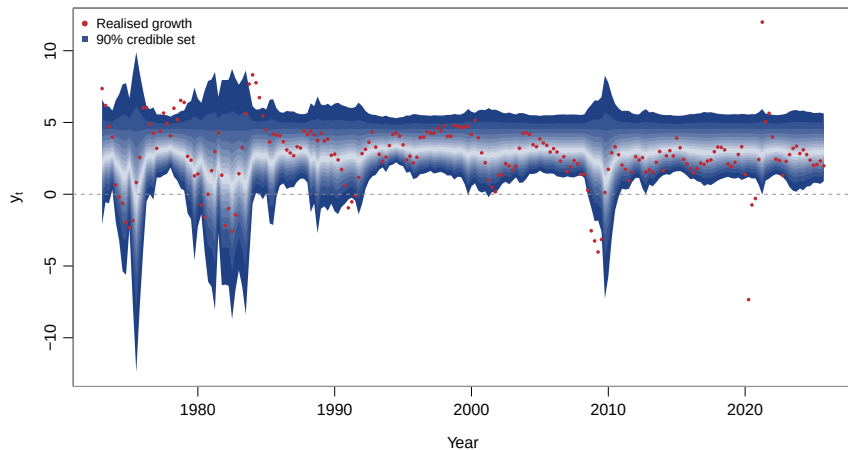
Against τ



Against NFCI



RECONSTRUCTED PREDICTIVE FIT FROM QRs



KEY TAKEAWAYS FROM SESSION 5

- **Asymmetric, time-varying tail risk:** Financial conditions load on the **lower** quantile of growth, barely the median, and the downside widens in recessions
- **Horizon matters** (Adrian et al. 2022, *AEJ: Macro*)
- **Robustness:** Out-of-sample the tail predictability is fragile (Plagborg-Møller et al. 2020, *Brookings Papers*); similar for the Euro area, Figueres & Jarociński 2020, *EL*)
- Multimodality rather than skewness (Mitchell, Poon & Zhu, 2024, *JAE*)?
- **ML & other extensions:** Bayesian QR with shrinkage helps (Carriero, Clark & Marcellino 2025, *JAE*); TVP-SV QR (Pfarrhofer 2022, *JEDC*); multi-country QR-FAVARs (Korobilis & Schröder 2025, *JoE*) and nonparametric/ML QR modelling (Clark et al. 2024, *JBES*)

Final digression: Tail risk forecast evaluation

TAIL RISK FORECAST EVALUATION

- › Standard CRPS weights all parts of the distribution equally (see Session 1)
- › For tail risk monitoring, we may want to emphasise accuracy in specific regions
- › **Quantile-weighted CRPS** (Gneiting & Ranjan 2011, *JBES*):

$$\text{qwCRPS}_t(w) = \int_0^1 w(\tau) \rho_\tau(y_{t+h} - Q_\tau) d\tau$$

where $w(\tau)$ is a weight function that emphasises selected quantile regions

- › $w(\tau) = \tau^2$: emphasises upper tail (upside risk, inflation)
- › $w(\tau) = (1 - \tau)^2$: emphasises lower tail (downside risk, GaR)
- › **Quantile score (QS)** directly evaluates the τ th quantile forecast:

$$QS_\tau = \rho_\tau(y_{t+h} - \hat{Q}_\tau)$$

TESTS FOR PREDICTIVE ACCURACY

- ▶ Diebold & Mariano (1995, *JBES*) test whether two forecasting models have equal predictive accuracy

$$d_t = S_t^{(1)} - S_t^{(2)}, \quad H_0 : \mathbb{E}[d_t] = 0$$

where $S_t^{(j)}$ is the score (e.g., CRPS) for model j at time t

- ▶ **Alternative to pairwise comparisons: Model confidence set** (Hansen, et al. 2011, *ECTA*) selects the set of best models given a pre-defined confidence level
- ▶ Can be applied to different predictive loss measures (e.g., CRPS, qwCRPS, or LPL)